

Time Walk: Blending Presence and History through AR Visualization

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Introduction

In anniversary events and exhibitions, visual timelines are commonly used as a means of guiding reflection on significant historical moments. However, these timelines often fail to foster a sense of connection among individuals who are unfamiliar with one another. To address this limitation, we propose utilizing AR to visualize a more immersive experience and enhance engagement with the timeline.

Prior AR visualization research [3, 4] predominantly focused on presenting historical events or scenes by effectively recreating and overlaying them onto present-day spaces or ruins. To better engage the visitors, many researchers [1] endeavor to enhance the richness of interaction through strategies like gamification and customization, enabling personalized augmented experiences. Nonetheless, a recent study [2] has underscored the importance of social presence in tour experiences, emphasizing its pivotal role in visitor retention and return rates. Thus, we aim to pioneer the visualization of user-generated data from AR product interactions to enhance engagement by showcasing social presence.

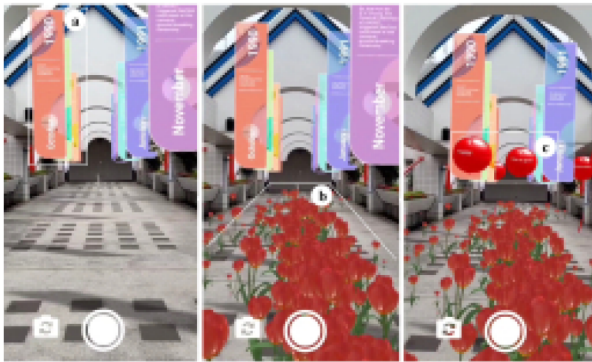
AR Visualization Design Elements

Walking among the timeline. Our first challenge was to design banners that conveyed information about a historical moment while evoking a strong sense of physical reality through the AR camera. To achieve this, we created glass banners adorned with colorful circles to depict a vibrant institute. The primary color utilized has an opacity of 100%, while the secondary colors have an opacity of 65%. In instances where the primary color intersects with a secondary color, the overlapping section is rendered with a lighter shade that incorporates white. The translucent nature of the glass material allows for a realistic representation in the AR camera, particularly when illuminated by physical light.

Our artwork enables individuals to physically traverse the timeline, providing them with a unique sensation of time-traveling and a nostalgic experience of “walking down memory lane.” This concept surpasses traditional 2D timeline designs, offering visitors an embodied and immersive encounter.

Sensing footprints. Our visualization artwork aims to foster a sense of connection and excitement among individuals across different times and spaces. To achieve this, we collected footprints left by the visitors while navigating the timeline and designed an artistic visual effect to convey the footprint data. By capturing the moments when visitors engaged with specific historical event banners and tracking their movement timestamps, we created an immersive experience. The artwork showcases blooming flowers at the visitor's footprint positions, inviting them to walk on a path adorned by the collective footprints of others, symbolizing the blessings of receiving more love and positive experiences.

As more people engage with the artwork, an increasing number of flowers bloom, mirroring the growth in communal connections. Similarly, the duration of one's presence in a particular spot is mirrored by the height of the flourishing flowers, further accentuating the presence of different visitors.



Leaving a voice on the timeline. We aimed to provide visitors with an avenue to express their emotions and thoughts regarding their involvement in these historical moments. To accomplish this goal, we implemented an interactive feature that allows visitors to upload their comments next to the historical event banners that they are browsing. During the formative design testing, we observed that many visitors expressed positive feedback (e.g., “Great!”) and some indicated their firsthand experience as witnesses (e.g., “I was there!”). To create a sense of collective presence and solidarity, we devised a captivating visualization technique that manifests the visitor's comments as floating comment balloons. These whimsical balloons gracefully ascend, carrying the visitor's remarks towards the sky, ultimately converging and harmonizing as a symbol of shared experiences and connections.

System

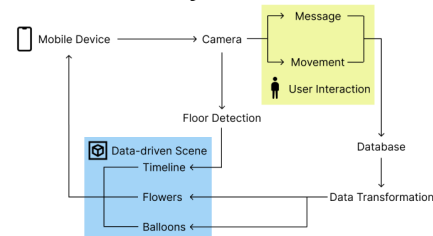
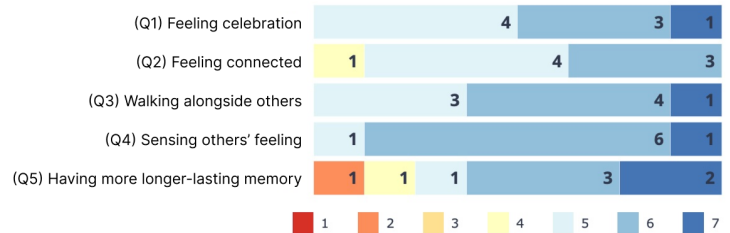


Figure 2 provides an overview of the sequential stages involved in the functioning of “Time Walk”. The diagram depicts a person at the center, utilizing a mobile phone camera to detect the floor, enabling the phone to visualize the historical moments’ timeline. The mobile phone records both messages and movement, which are subsequently stored in a database. “Time Walk” processes this data, converting movement into trails of flowers and messages into balloons. As more individuals engage with “Time Walk”, the data accumulates, leading to an aggregation of experiences.

User Study and Results



To conduct an initial assessment of our designs’ effectiveness, we conducted a pilot study involving eight participants affiliated with the university (four males and four females from 23 to 29 years old; P1 - P8). They were invited to experience our artwork installation, complete a survey, and provide feedback on the three design elements.

Notably, all users rated Q1 five points and above, and rated Q2 four points and above, indicating a strong sense of celebration and connection with unfamiliar individuals during the anniversary event. Participants described their experiences as “romantic” and “immersive.” All users rated Q3 and Q4 five points and above, showing their appreciation for the flower and balloon designs. Specifically, P7 compared AR and real-life experiences, expressing enjoyment and a sense of safety while interacting with floating balloons through AR technology. Furthermore, seven users agreed that the immersive timeline would be more memorable than a general timeline (Q5). P2 wrote, “The colorful banners reflected the colorful historical events of our university and made me memorize the events.”

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Conclusion and Future Work

In summary, our research introduces “Time Walk”, an AR visualization artwork that offers interactive experiences through timeline navigation, footprint sensing, and message leaving. We crafted three design elements that enhance realism and foster a sense of social presence. By leveraging mobile phone cameras, our implementation effectively visualizes the timeline while processing user movements and messages. The user study results demonstrate positive feedback, emphasizing the celebratory atmosphere, immersive nature, and appreciation of design elements. Moving forward, we aim to expand the range of interactive elements by enabling customization options and conducting larger-scale studies to further enhance user experiences in interactive AR visualization artworks.

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